Color, Texture, Animation, and Sound

4

Our ability to sense color has adaptive value. For example, it helps us know when fruit is ripe. It's a fact that people who have lost the ability to see color (following brain damage) report that their perception of food has changed dramatically (for example, one such person described a tomato as "black and repulsive").

In general, color helps us notice what's different and what's the same; we easily see clusters of ripe red apples against the green of the tree's leaves, or a red sign mounted on a black wall. These functions of color perception in everyday life don't go away when we see PowerPoint presentations; vision operates the same way—for better or worse—regardless of whether we are viewing apple trees or PowerPoint slides.

In addition to color, almost all information graphics software uses textures as filling (both in the foreground and the background), and many PowerPoint presentations use animation (i.e., movements on the screen) and sound—both within a slide and in transitions between slides. Again, the cognitive mechanisms that pick up and process such information during everyday life don't go away when we attend a PowerPoint presentation; they operate the same way, which can either help or hurt the presentation. In this chapter I summarize recommendations for using color, texture, animation, and sound.

Color: The 12 Essentials

Color not only adds visual interest to a display but also can help you to communicate effectively. If used improperly, however, variations in color can obscure your message. To help you understand how to use color to make your presentations eye-catching and compelling, I first need to explain some essential facts about color. If you understand why I make my recommendations, you won't need to memorize them.

Color is composed of three distinct aspects:

1. Light can be thought of as a series of waves, similar to regularly spaced waves on the surface of the ocean. The longer the time between the arrival of each peak of the waves, the greater the wavelength. And different wavelengths of light are perceived as having a different *hue* (what we usually mean by color, its qualitative aspect—red, green, etc.). We see very long wavelengths as red and very short ones as violet, with all other hues in between.

   Here's a particularly important fact: We don't see decreases in wavelength as a smooth shift in hue. Rather than being arranged along a single scale (for example, as is loudness, where increasingly greater amplitude produces increasingly louder sound), hue is psychologically arranged around a circle—the famous "color wheel." Some colors are nearby,

others are opposite each other. Thus, it's not simple or straightforward to use variations in hue to represent different quantities; changes in wavelength don't psychologically correspond to the impression of greater or smaller quantities.

2. The peaks of the light waves can be relatively high or low, which corresponds to the amount of light. That is called the lightness when the light is reflected (if the display is projected onto a screen) or brightness when the light is emitted (if the display is shown on a monitor). In either case, the amount of light can be varied by the amount of gray that is added. Hereafter, I will use the term "lightness" to refer to the amount of light in the display, on the assumption that most presentations will be projected, but the recommendations apply equally well to cases when presentations are shown on a monitor.
3. The purity of the color is its *saturation,* seen as the deepness of the color, which can be varied by the amount of white that is added. Adding white has the effect of "washing out" the hue, making it increasingly more subtle (such as in pastel colors). Saturation can be varied separately from lightness (as you can see on a monitor, when you turn up the intensity). And, unlike hue, both the lightness and saturation of a color can be varied quantitatively, producing the psychological impression that "more is more."

   Color perception is complex in part because the perception of hue depends to some extent on the surrounding colors, and so the display must be looked at as a whole. Furthermore, people vary greatly in their perception of color. A sizable percentage of the population (about 8 percent of American males and .5 percent of American females) is color-blind—that is, they have trouble distinguishing certain colors, typically red versus green. In addition, people vary greatly in their assessments of how intense two colors must be to have equal lightness.

   The following recommendations will help you to ensure that the colors you use will be discriminable by everyone, that they don't irritate the viewer, and that they convey information properly.

1. Use colors that are well separated in the spectrum.

To increase discriminability, colors that are included in your display should be separated by at least one other noticeably distinct color in the spectrum, depicted in the color wheel. The six colors that we perceive as being most separated are *reddish-purple, blue, yellowish-gray, yellowish-green, red*, and *bluish-gray*. The "11 colors that are

never confused" are *white, gray, black, red, green, yellow, blue, pink, brown, orange,* and *purple*.

But my strong advice is never to use all 11 in a single display: To do so would make the word "garish" seem inappropriately limp. Aesthetics aside, even when short-term memory limitations are not an issue (because the colors all are physically present), because in general people can easily keep in mind the differences among only nine colors.

2. Avoid using red and blue or red and green in adjacent regions.

In spite of the requirement that colors be well separated, and hence easily discriminable, you should not use hues that arise from very different wavelengths to define boundaries. The lens of the eye, unlike lenses in good cameras, cannot properly bring into focus two very different wavelengths at the same time. This is why red (a long wavelength) and blue (a relatively short wavelength) will seem to shimmer if juxtaposed. For most purposes, this effect will distract the viewer and should be avoided.

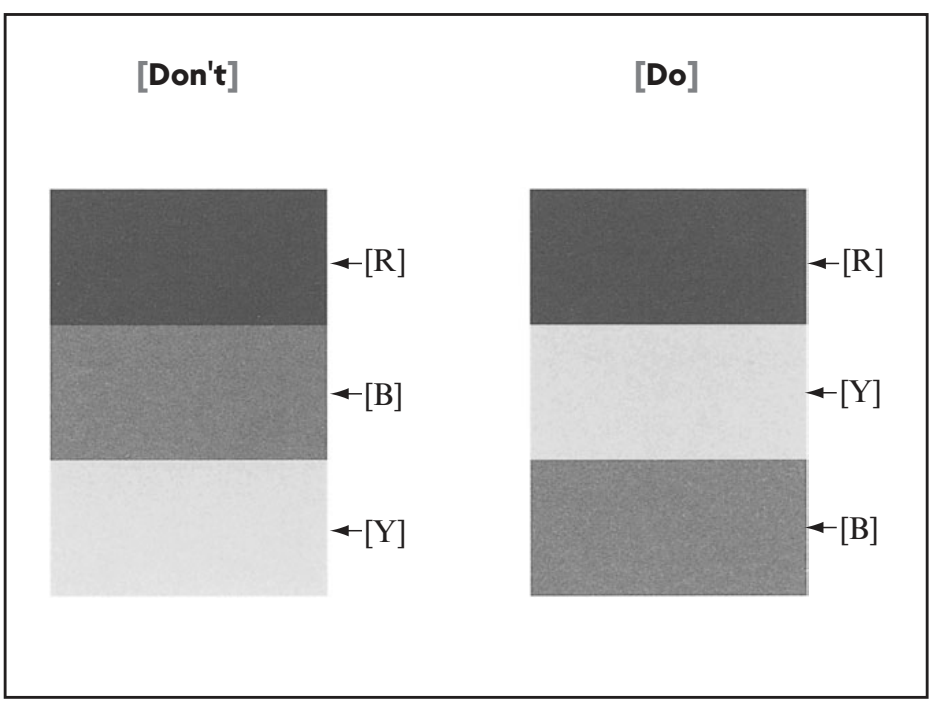


Figure 4.1. The eye cannot properly focus on red and blue at the same time, so a boundary should not be defined by juxtaposing red and blue regions. "R" is for red, "B" for blue, and "Y" for yellow. (See this figure in the color insert.)

In addition, it is usually a good idea to avoid using red and green to define a boundary, because a sizable percentage of the population has trouble distinguishing these colors (as noted earlier, this is the most common form of color-blindness).

3. Make adjacent colors have different lightnesses.

Our visual systems have difficulty registering a boundary that is defined by two colors that have the same lightness. I discovered this the hard way while driving one winter dusk. A stop sign had been posted on a green steel girder that was holding up an overpass. The light was just right (i.e., wrong), and I didn't distinguish the red from the green—and was promptly pulled over and ticketed by a waiting patrolman. I decided that it really wasn't my fault; it was my all-too-human visual system's fault. So I went in to a preliminary hearing, to explain this to the magistrate. But seeing the look on his face after my mini-lecture, I gave up and paid the fine—and decided to be extra vigilant during dusk.

Unfortunately, you can't predict exactly how great the differences in lightness must be to discriminate boundaries between different hues easily. But, all else being equal, this may help: When colors reflect the same objective amount of light, we see blue as the lightest (or brightest) color, followed by red, green, yellow, and white. To be on the safe side, I recommend adjusting lightness subjectively, ensuring that it is obviously different in adjacent colors, for two reasons:

1. People vary considerably in their perception of how light a color is, and this can vary for different colors.
2. Different sorts of room lighting affect different colors in different ways.

4. Ensure that the foreground and background are discriminable.

The 19th-century British genius Charles Babbage is credited with inventing the first calculating machine. Less well known is the fact that Babbage took extraordinary care before he published his table of logarithms in 1826, to the extent of testing the ease of reading 10 different colors of ink on 50 different colors of paper. The winner? The blackest ink on a light buff paper. In fact, the most discriminable colors are black and white. If you want to use these colors (perhaps because you are speaking in a huge room and want to ensure that even those at the very back can read your slides), consider how the room will be lit.

- If the room will be well lit, you are better off using black figures on a white background.
- If the room will be almost dark, a white background will appear very bright, perhaps even irritating (even if a light is shining on you, which I recommend). So, in this situation you generally are better off using white figures on a black background.

However, black and white are not the most exciting colors, and you won't lose much by using different hues for figure and background as long as they are clearly discriminable. To ensure that they are easily discriminable, be sure that the colors differ obviously in terms of all three qualities: hue, lightness, and saturation.

5. Make the most important content element the most salient.

Larger color differences will be noticed first, so ensure that the most important object or segment stands out the most.

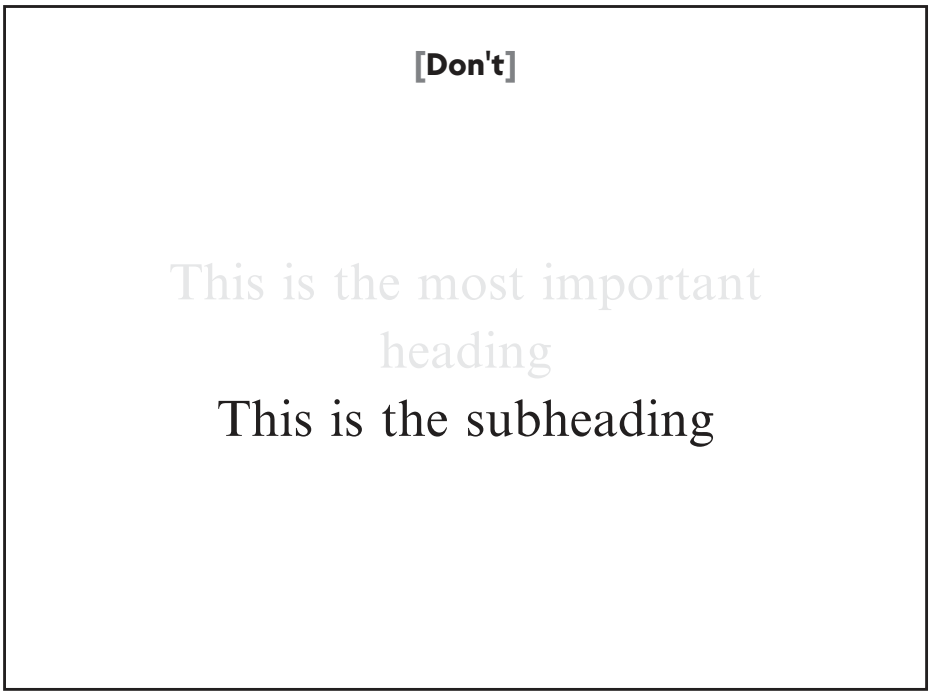
- If no particular element is most important, make all elements equally salient.
- To make elements equally salient, after you adjust the hues and lightnesses so that the colors will be easily distinguished, adjust the saturations for the different hues you are using until none dominates.

6. Use one color for titles and another, less salient one for text.

To help the viewer organize your text, use one color (e.g., yellow-orange on a dark gray background) for titles and another color (e.g., white) for text entries beneath the title. But, as noted in Chapter 3, ensure that the title is a more salient color than the text, thereby appropriately directing the viewers' attention.

7. Use warm colors to define a foreground.

This recommendation is based on nonintuitive discoveries about visual perception. The brain senses depth in part by exploiting the slightly different images that are registered by each eye (because they are located in different places on the head). By analogy to a similar process in hearing, this is called *stereo vision*. Stereo vision works



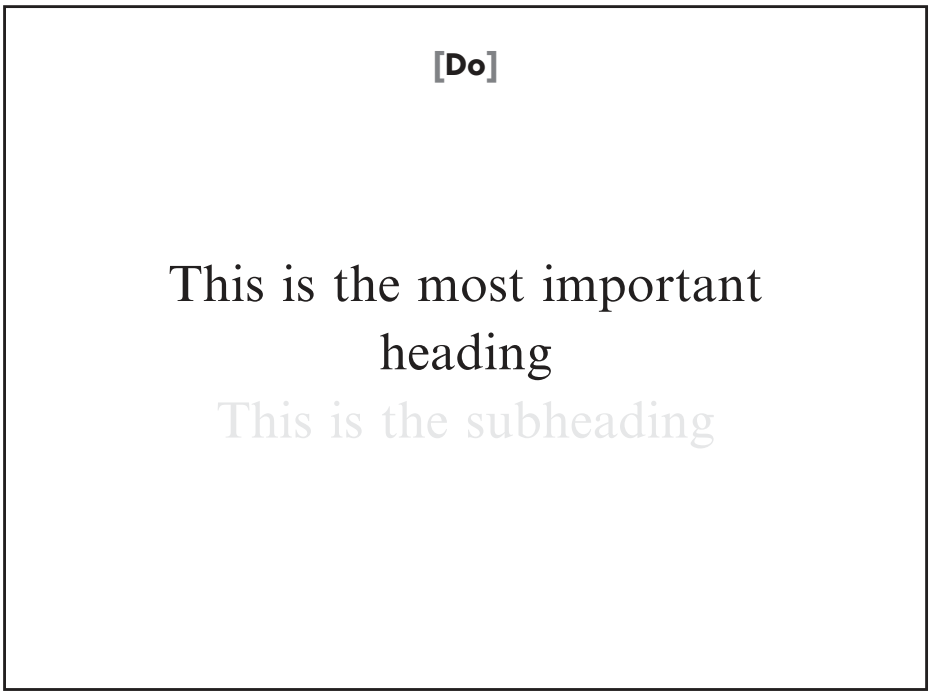


Figure 4.3. The salience of colors should reflect the importance of the headings. (See the insert for the color versions of these figures.)

well to specify distance (up to about 10 feet away from the viewer), but it can be tricked to produce an interesting illusion. The illusion occurs because light waves usually contain a mixture of different wavelengths, which we see as different hues. You can separate out the different hues in ordinary light by holding a prism in front of a window and observing the rainbow that is projected on the wall. The lens of the eye acts like a prism because the eye is aimed slightly toward the center when we look at objects in front of us, and so the lens is angled. Light of different wavelengths is projected to slightly different locations on the back of the eye just as it is projected to slightly different locations on the wall when we hold up a prism. And this effect differs slightly for the two eyes, which produces a "false stereo" effect. The result: A "warm" color, such as red, yellow or orange, will appear to be in front of a "cooler" one, such as green, blue, or violet.

This odd fact about visual perception leads me to recommend that text should be a warmer color than the background, so that the background doesn't seem to fight to move in front of the text. Moreover, when two lines cross (as in Figure 4.4), the warmer one should pass over the cooler one; if it does not, the back line will seem to be struggling to come forward, trying to snake around the one in front. As entertaining as such effects may be, a good presentation is no place for a visual wrestling match!

Figure 4.4. A red line that appears to be struggling to move to the foreground produces an effect that is neither esthetically nor functionally desirable (see this figure in the color insert).

8. Don't use deep, heavily saturated blue for text or graphics.

Avoid rendering text or graphics in a deep, heavily saturated blue; the eye cannot focus the image properly, and so deep blues will appear blurred around the edges.

- Similarly, avoid cobalt blue, which is in fact a mixture of blue and red; for people with normal vision, this color can never be fully in focus because the eye cannot accommodate to both wavelengths at the same time. The halo you have probably noticed around blue street lights at night is not fog; your eyes are failing to focus the image properly.

9. Use color to group elements.

Regions of the same color will be seen as a group. Use the same color for all titles and another color for all text entries, which will clearly group the material into these two categories. And use color to pair corresponding elements, as in Figure 4.5.

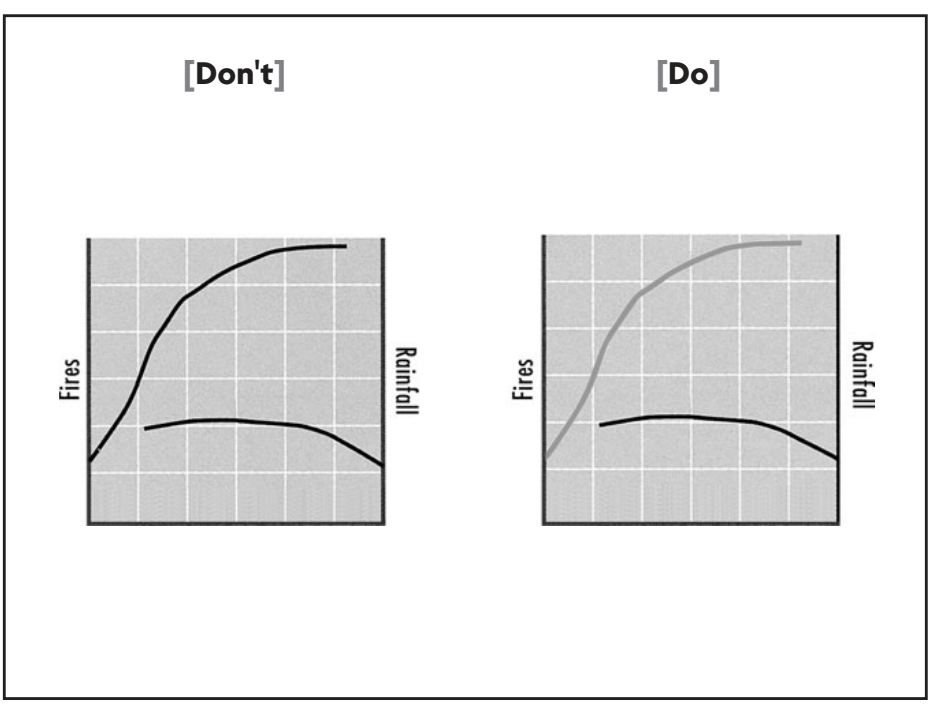


Figure 4.5. If used properly, color can be a very effective grouping device. In this example, color in the right panel groups the appropriate scale with the corresponding line (see this figure in the color insert).

Example: Grouping by similarity is very useful if you want the viewers to compare two or more elements in different places. For instance, if two pie graphs are used, using the same color for corresponding wedges will group them effectively.

10. Increase saturation and lightness for hues that indicate greater amounts.

Don't vary hue by itself to represent differences in amounts. Using hue to represent amounts requires the viewer to memorize a key because hue itself does not vary along a continuum of less-to-more. For example, shifting from red to violet does not convey the impression that an amount has been added the way that shifting from a short bar to a tall bar does. Although the wavelengths of the hues order them into red, orange, yellow, green, blue, indigo, and violet, hues are not a psychological continuum.

In some situations you may need to use hue to convey quantitative information, such as occurs in pictures of the brain that indicate the amount of activation when a person performs some task, or pictures of a model airplane in a wind tunnel that indicate the relative friction along different surfaces. In such cases, color is used to specify quantities associated with each portion of an object.

- For such graphics, use deeper saturations (more color) and greater lightnesses (more light) for hues that indicate greater amounts. We see increases in both of these visual dimensions as increases in amount, and so these increases can signal increasing quantities effectively.
- If you must use hue to convey variations in quantity (e.g., to show variations in temperature across a complex surface), use psychologically equal changes in color to convey equal increments in the amount—and you'll need to experiment to get this approximately right. If the psychological increments don't seem about equal, viewers will assume that more similar quantities are conveyed by more similar colors.

11. Don't use hue, lightness, and saturation to specify different measurements.

The recommendation to co-vary the three aspects of color to indicate greater amounts will also lead you to avoid using hue (the qualitative aspect of color), lightness (the amount of light), and saturation (the deepness) to convey different types of information.

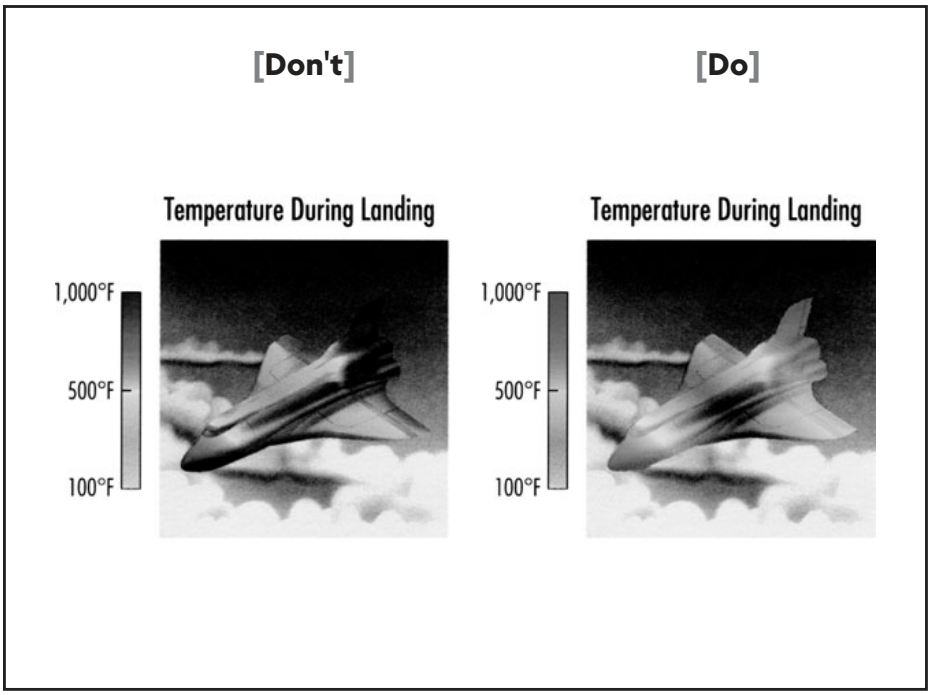


Figure 4.6. Because differences in hue are not immediately perceived as differences in amount, the viewer is required to memorize a key if hue alone is used to represent quantities (as in the version on the left). However, if the saturation and lightness vary along with hue, as in the version on the right, the viewer can easily see differences in represented amounts (see this figure in the color insert).

- Don't be tempted to specify three different types of measurements in a single graphic. The viewer will have a very difficult time sorting them out because hue and lightness, and hue and saturation, are so-called integral perceptual dimensions (they group together perceptually; see the discussion of the Principle of Perceptual Organization in the appendix). A viewer looking at a colored region cannot help but pay attention to its depth and lightness as well as its hue.

12. Respect compatibility and conventions of color.

A friend of mine was teaching a seminar on the cultural revolution in China. I asked him what color he was using for the background in his PowerPoint slides, and he said, "Red." When I asked him to describe the color in more detail, he said that it in fact was more like a burgundy. This was a wise choice, for two reasons:

1. Fire-engine red is a warm color and will be seen as closer than cooler colors—not a good idea for a background.

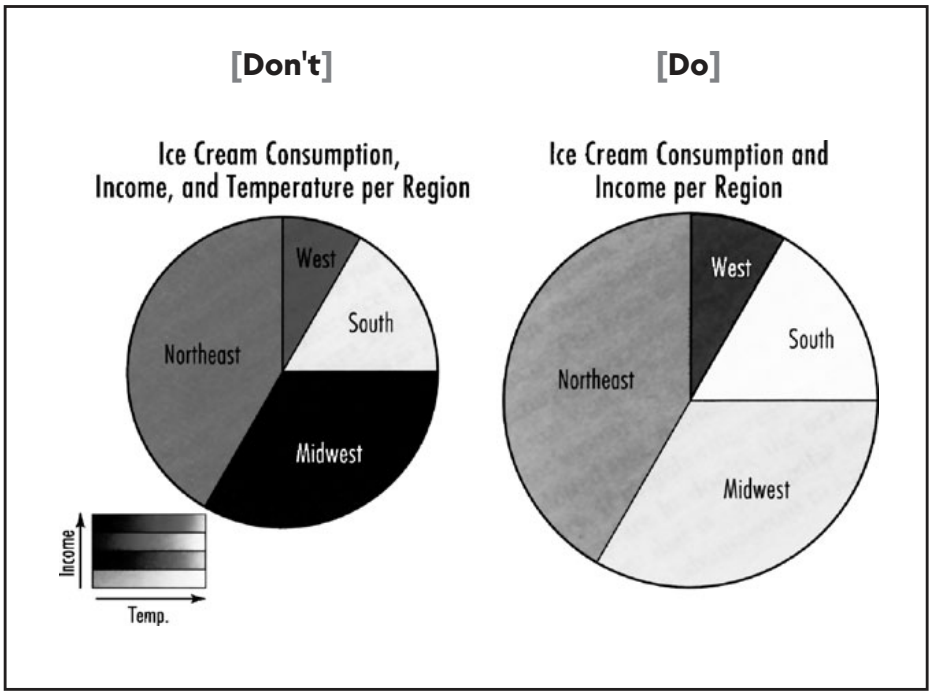


Figure 4.7. The display on the left uses size to indicate consumption by region, hue to indicate income, and saturation to indicate temperature; it is a puzzle to be solved. The display on the right uses only size and saturation to convey amounts, and viewers can clearly sense the orderings here (see this figure in the color insert).

2. However, using a shade of red is compatible with the topic, red being the color of the Chinese revolution.

Objects and events sometimes have characteristic colors, different cultures have specific conventional color symbols, and colors may also have political connotations, for example, in the United States, "red states" are conservative, and "blue states" are liberal. Although the colors probably were not intended to convey specific connotations, we nevertheless can ask what red and blue signify in the United States. The best survey I know of on this topic was conducted by Joe Hallock (http://joehallock.com/edu/COM498/credits. html). Here are some of his survey results, indicating the percentage of people who associated the colors with the concepts (for example, he asked them to indicate which color "best represents trust"). I've focused here just on blue and red:

Concept	Blue	Red
Trust:	34	6
Security:	28	9
Speed:	1	76
High quality:	20	3
Reliability/dependability:	43	3
Fear/terror:	0	41
Fun:	5	16
Favorite color:	42	8
Least favorite:	0	1

In short, for Americans, blue is generally a more desirable color. But even this depends on the specific message you want to present; if you want to convey speed, fear, or fun, red would be more effective.

Texture: 2 Points Regarding Hatching

Individual bars in bar graphs, wedges of pie graphs, regions of maps, or portions of diagrams are often filled with a textured pattern. The following two recommendations will help you make helpful hatchings.

1. Ensure that textures are discriminable.

As discussed, if neighboring regions are too close in color, the brain will not easily establish a border between them. (That's why a polar bear lying on the snow is hard to see. Camouflage is advantageous to a polar bear, not to a display.) By the same token, if two regions differ only by their texture, the brain will not easily establish a border between them if the textures are too similar. Here's how to ensure that textured regions are easily discriminable:

- *Vary orientation by at least 30 degrees of tilt*. When you read an analog clock (one with hands, not just numbers), some positions of the hands require closer attention than others. In accordance with the Principle of Perceptual Organization (specifically, the aspect of Input Channels; see the appendix for details), if differently oriented hatchings are used to distinguish regions, their tilts should differ by at least 30 degrees (the angle formed by the hands of a clock when they point to adjacent numbers, as at 12:05; see Figure 4.8).
- *Vary the spacing of texture patterns with similar orientations by at least a ratio of 2 to 1*. If cross-hatching, stripes, dots, dashes, or other regular patterns have similar orientations (that is, within 30

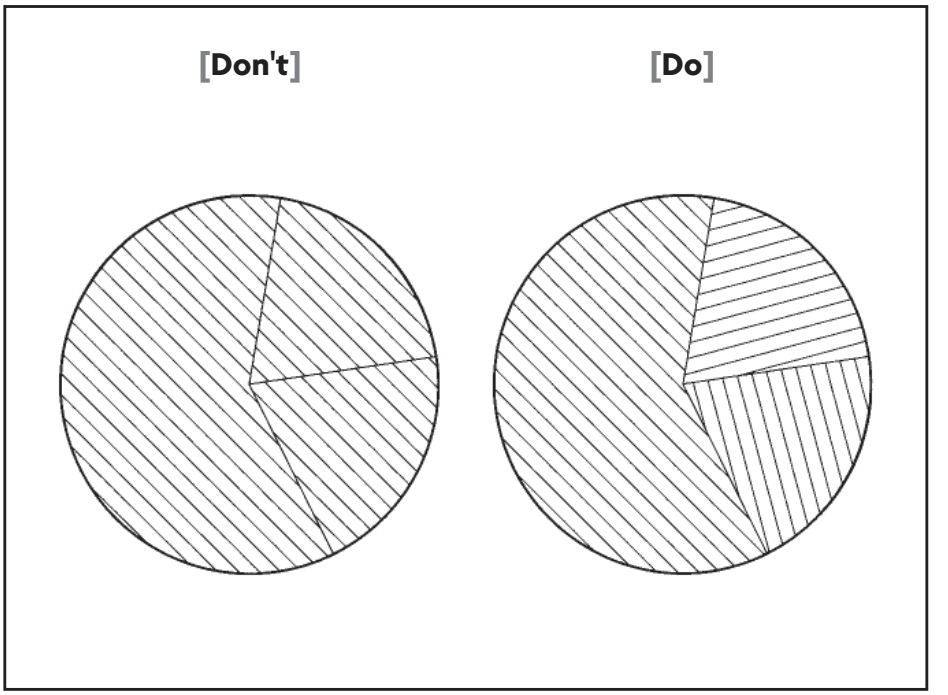


Figure 4.8. Line orientations must be immediately discriminable to delineate regions clearly.

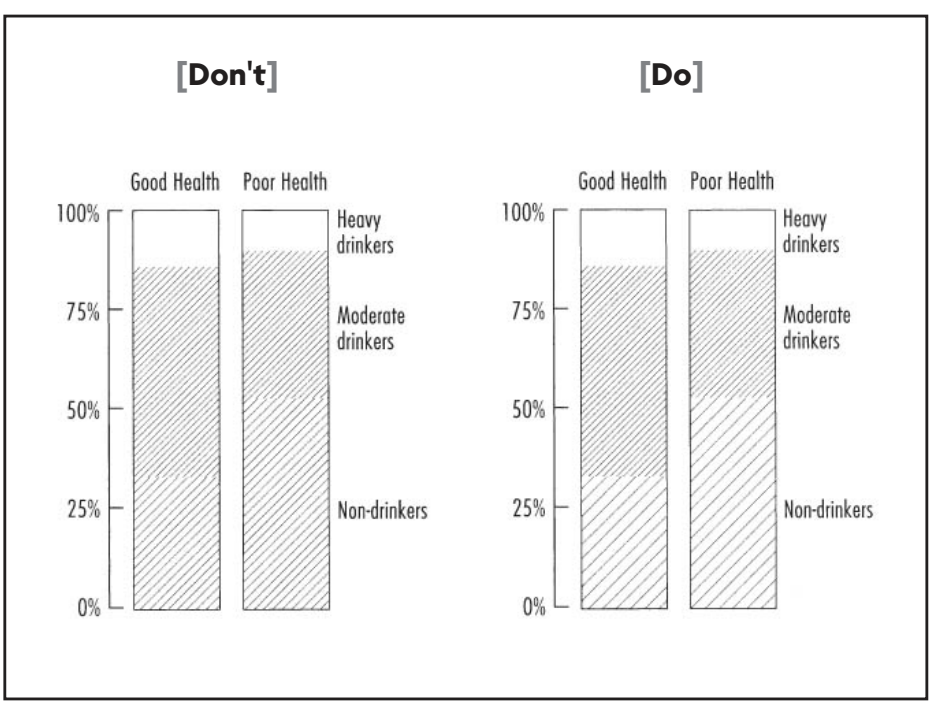


Figure 4.9. Line spacings that differ by at least a 2:1 ratio are immediately discriminable; if regions are not immediately discriminable, the viewer has to work harder than necessary to compare corresponding elements.

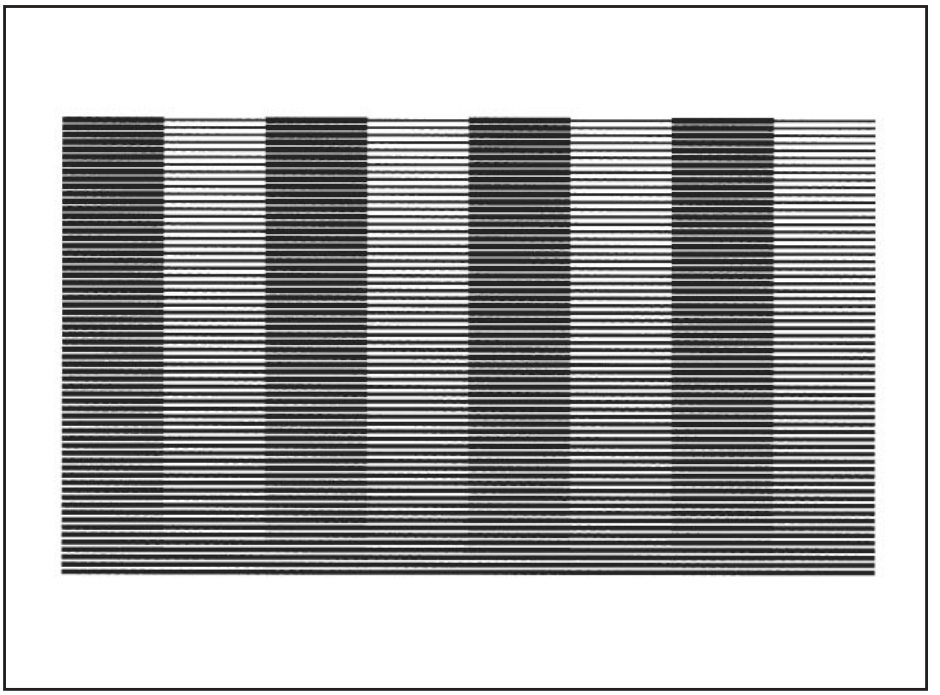


Figure 4.10. An annoying shimmer occurs when your visual system is struggling to detect a poorly defined edge.

degrees of each other), the densities of the pattern should differ by at least 2-to-1 (Figure 4.9). If a region has 8 hatch lines to the inch, to be immediately discriminable adjacent regions should have either 4 or fewer lines to the inch, or 16 or more lines to the inch.

2. Avoid visual beats.

Similarly oriented patterns of stripes or texture elements may appear to shimmer, a distracting and irritating effect (Figure 4.10). This effect is less likely to happen if the stripes of texture elements in different regions are easily discriminable, as specified in the previous recommendation.

Filling: 3 Points about Background Patterns

In many PowerPoint displays, the background contains a pattern, which is included primarily to make the display more attractive or interesting. If you choose to use a pattern for your background, such as those provided in the PowerPoint templates, be careful to ensure that it does not obscure the message.

1. Ensure that the background pattern is not salient.

Every once in a while I see a movie with an actor so talented that a minor character "steals the show" (think of Bill Nighy's role as Billy Mack in *Love Actually*). Don't let this happen to your presentation, where the background is so interesting that it steals the show. The background should not distract from the information-bearing lines and regions, nor should its salience lure the eye away from these elements. A background should be lighter and in less saturated colors than those used in the rest of the display, have few details, and have soft edges.

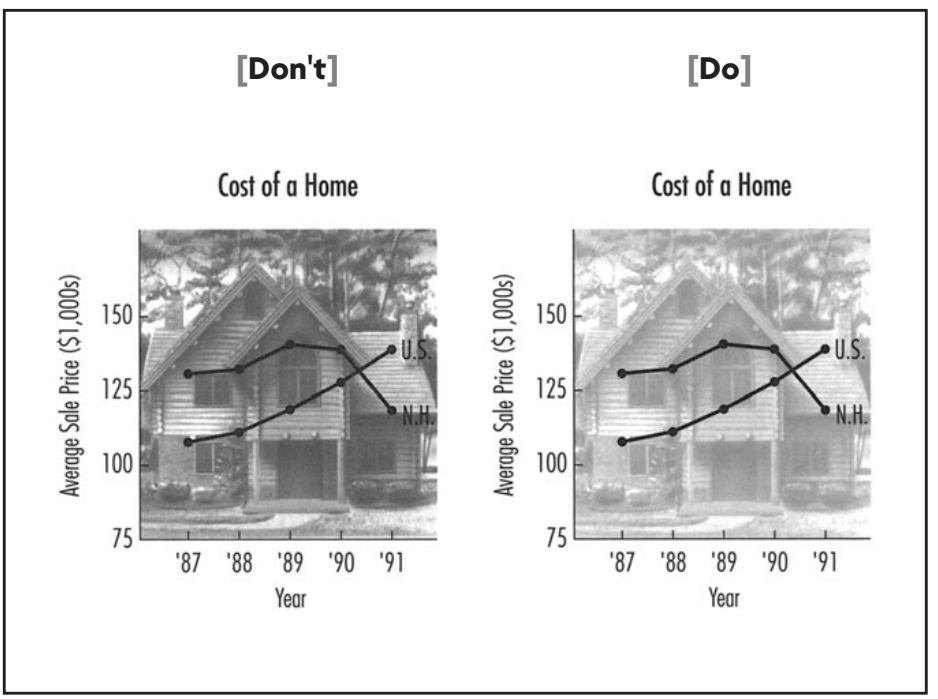


Figure 4.11. If salience leads readers to see the house before the data, they will have to work to sort out the content.

2. Don't allow parts of background patterns to group with parts of the foreground.

Use contrasting hue, saturation, and brightness to ensure that background figures don't group with parts of the display itself, producing confusion between background figures and content (Figure 4.12).

3. Use a background pattern to reinforce the main point of the presentation.

An effective background can underline (as opposed to undermine) the overarching message of the presentation, providing a kind of pictorial

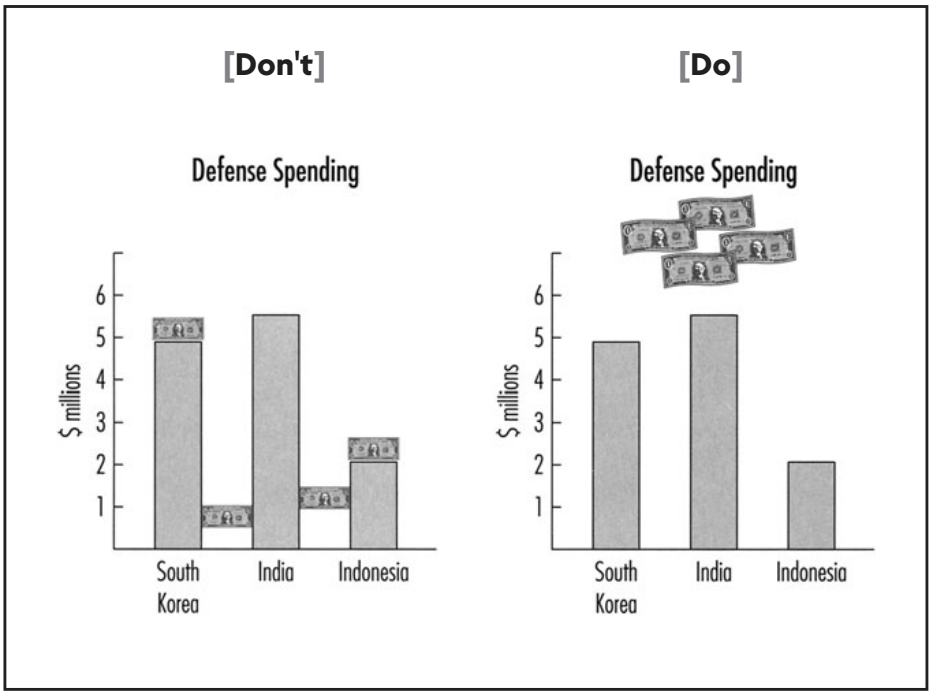


Figure 4.12. If background elements group with content, the display is likely to be misinterpreted.

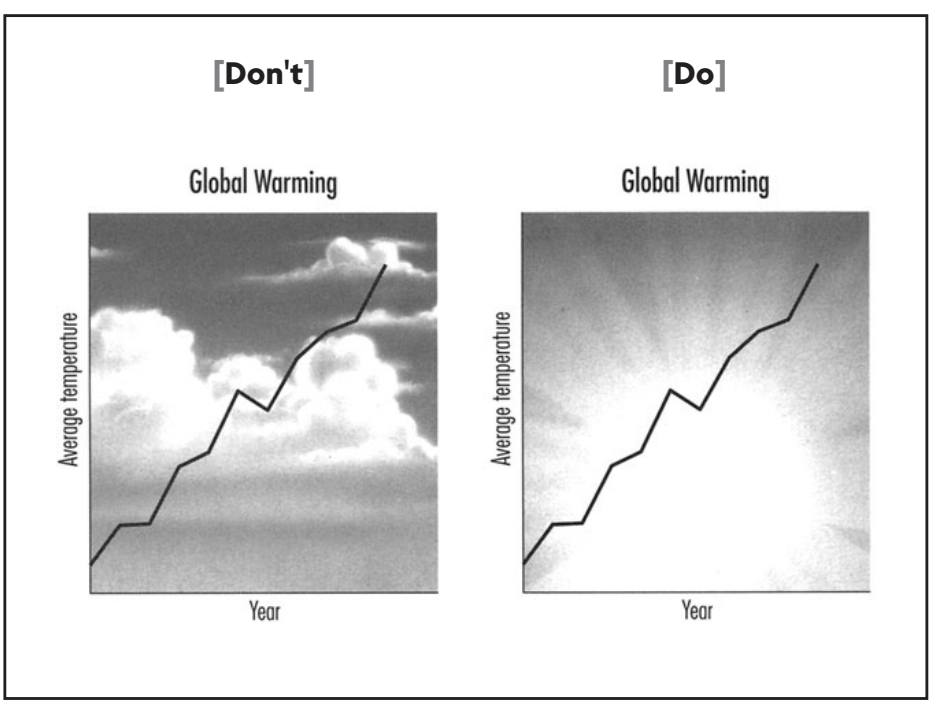


Figure 4.13. The editorial content of a background can allow you to underline your message effectively, or it can create confusion; the background pattern should not conflict with the message of the display.

label (Figure 4.13). In order to be effective, the background design must be compatible with the content of the presentation. To get a quick sense of whether you are on track, show a couple of friends the background image that you are considering using and ask them to describe it; this description should be appropriate for the topic and should have connotations that are compatible with your message.

- Every element of a display should convey information (such as the fact that amounts are specified in U.S. dollars in Figure 4.12). Tacked-on elements—such as pictures of the planets or swirls of different colors—will only distract the viewers.

Animation and Transitions: 10 Recommendations

Recently, I experienced one of the most disheartening things that can happen during a presentation: One of the audience members seated near the front of the room stood up and left during the middle of my presentation. Her standing up and then walking the length of the aisle to the exit was not only an apparent comment on my presentation (she was "voting with her feet"), but it also caught the attention of the rest of the audience. I had to speak a bit more loudly, and be a bit more animated for a moment, or I would have lost even more of the audience as their attention was distracted. But it wasn't their fault; the human brain is wired so that change—including change in position—automatically catches our attention. This makes animation (i.e., movement of a pattern on a screen) a two-edged sword: It can be used to direct the viewers' attention to what you want to say, or it can distract them from your message. My recommendations focus on ways to ensure that animation does in fact work for you, not against you.

1. Use animation to direct attention.

If you are discussing a sequence of events that you've represented graphically, use animation to guide the viewers sequentially through the graphic.

Example: If you present a flowchart, have the arrows appear only when the transition from one phase to another becomes relevant, and have the arrows literally grow from the process (box in the chart) just being discussed to the next one.

- Even if you are not discussing a sequence of events, you can use motion to guide attention: In general, presenting a label first and then having an arrow extend from the label until it touches the labeled part is an effective way to direct the viewers' attention.

2. Use novel transitions to catch attention.

The PowerPoint program provides many options for transitions between slides, such as by having a slide enter from the left or by having it built up from fragments that come together. The first time you use a transition, the novelty will grab the viewers' attention. Change of any sort should carry information—and such change does: The information being "Wake up and pay attention! This is especially important!" But after that, the audience will adapt and come to expect that transition—and a new type of transition would be necessary to grab their attention later, to emphasize yet something else that's important. I would not recommend making such changes more than once or twice during your presentation, however; people adapt quickly, and what changes can soon become what's staying the same. As the French put it, *plus ça change, plus c'est la même chose*—the more things change, the more they stay the same.

3. Don't randomly use different transitions for different slides.

Use the same transitions throughout the presentation, unless you want to emphasize a particular slide (as noted above). Don't change the type of transition randomly.

4. In general, don't use slow fade-in or fade-out.

The PowerPoint program allows you to make a slide fade in or fade out. If this is rapid (less than 1 second), it is simply another form of transition. But if a slide slowly becomes visible (taking more than 1 second to appear fully), the viewers first will see parts that are not immediately discriminable, which will require either that they work to read the slide or that they wait until it is clear. Neither situation is ideal, and you risk allowing the viewers' attention to wander.

Some presenters use a slow fade to give viewers a moment or two to absorb complex information. However, in this situation I recommend simply leaving the slide visible until you think they have understood the material.

5. Don't move portions of the same text line or graphic separately.

Move portions of the same piece of information (text or graphic) in the same way, so that they group together.

Example: If each bulleted item in a list enters from the left, ensure that all words in an entry come in together. Having them arrive one at a time, or from different locations, only makes the viewer have to work in order to organize them into an interpretable unit.

6. Don't move more than four separate perceptual groups simultaneously.

To avoid overloading our ability to track and hold in mind information, don't have more than four groups (defined by the Principle of Perceptual Organization) moving at the same time.

7. Don't require viewers to read moving words.

Ask anyone who has ever stood in Times Square in New York City and tried to read the news on the ticker-tape display: Moving words are hard to track and interpret. Any gain in visual interest is more than canceled by the added processing difficulty.

8. Build up complex displays a part at a time.

If you are showing a complex display, you can help the viewers by using animation to build it up, a part at a time (while keeping the portions of each part together, as already noted). Having a part enter from the top ensures that the viewers will focus on just that new part, and then see how it is integrated into the whole.

Example: If you were showing a map of the southern states and wanted to indicate average weekly coffee consumption per capita, you might use icons of stacks of coffee cups to indicate the amount (each cup would signify a certain amount of consumption, ideally one cup). Each stack would be built up as coffee cups dropped down from the top, one at a time onto a specific state, with the final height indicating the average amount of coffee consumed. If the precise amount were important, you could have the numerical value appear at the top of the stack

after the cups had slid down to their proper resting places. Using animation in this way will direct the viewers' attention to just the material you want to discuss at any one point.

9. Make animations compatible with the represented object or event.

Many objects and events have characteristic movements, which should be preserved in the way you have them move on the screen.

Example: If you show a picture of an automobile, don't have it rise up from the bottom of the screen or seem to drop down from the sky. Have it enter from the left if it faces to the right, or from the right if it faces to the left. Similarly, if you want to use the height of a flower to indicate the amount of rainfall, have the flower grow vertically.

10. Use videoclips to illustrate a relevant event.

When integrated into a PowerPoint presentation, video clips can illustrate an event that unfolds over time better than a description or set of still slides. All of the recommendations regarding Clipart in the following chapter also apply to video clips. In addition, a word to the wise: Be sure that your video will in fact operate on the computer you are using—videoclips are memory-intensive and will not run on all machines.

Sound: 10 Ways to Avoid Making Noise

The PowerPoint program allows you to present sounds, either from a built-in menu of choices (which includes sounds such as a dog barking or a man laughing) or custom-defined. The following 10 recommendations apply to the use of sounds.

1. Use sounds to grab the audience's attention.

An unexpected or distinctive sound typically has the same effect as a sudden motion: It grabs the audience members' attention. However, unlike motion, sounds don't inherently direct the audience to a specific item on a slide. Rather, sounds alert the audience to another

event, such as a change in text or graphics. If a slide has been visible for a few minutes, and only a portion of the slide is then changed, an appropriate sound will help to draw the viewers' attention to the change.

Example: If you want to show how lightning can spark a forest fire, you may first need to show the audience a picture of a forest and describe the relevant conditions. If this takes you a few minutes, you might then want to present an audible "crack" when you show the lightning bolt.

2. Use sounds sparingly as alerts.

If you use sound to grab attention, do so sparingly. The first time you do this, it will wake up the audience and alert them to the upcoming event. By the third time you do this, the audience may become annoyed—and may even begin to filter out the sound.

3. Use sounds to define the context.

Sounds can set the stage for your entire presentation, a portion of your presentation, or a specific point. These sounds should be very transitory, however, functioning as a kind of auditory title.

Example: If you are talking about pollution in the seas, you can start with the sound of surf. Or if you are talking about the growing barter economy, you can play the sound of an open-air market. Both National Public Radio (NPR) and the British Broadcasting Corporation (BBC) radio news make very effective use of such sounds to set the stage for their feature stories.

4. Use sounds that are appropriate for the topic and point being made.

If you use a sound to introduce a slide, ensure that it is compatible with the point.

Example: If you want to introduce a presentation or section on the effects of a hurricane, the sounds of a ferocious wind might be appropriate, especially if accompanied by a video clip of the wind ripping through a town. The sounds of gentle rain on the roof would not be appropriate.

- If you use an inappropriate sound, it will seem funny—which is not always a bad thing, but you should intend this effect, not fall victim to it.
- My advice: Never, ever use rude or crude sounds (you know what they are).

5. Use sounds to allow the audience to "come up for air".

Sounds can serve as a kind of punctuation in a complex presentation (again, NPR and BBC radio are very good at doing this); giving the audience a chance to "come up for air" allows them to process the information you've just presented.

Example: If you've been presenting data on emerging stock markets in the former Soviet Union, you might have a few seconds of the sound of the floor of one of the exchanges, accompanying an appropriate title on a slide, such as "Russian Stock Exchanges."

- However, you should use sound in this way only very sparingly, and your timing must be just right. Otherwise you will not seem serious, and the audience will discount your message.

6. Use sounds to provide evidence.

In spite of the ample evidence to the contrary (served up daily by creative artists armed with computer graphics programs), for most people seeing is believing. The same goes for many sorts of sounds. Sounds, especially recorded dialogues, can serve the same roles as photographs when used as evidence.

Example: If you want to illustrate typical interactions between physicians and patients, brief sound clips that provide samples of such interactions might be appropriate.

7. Coordinate sounds, text, and graphics.

Although you can show a blank background slide when you present sounds, doing so misses a valuable opportunity to provide the audience with dual means of learning the material, via sight and sound. It's best to present sounds with appropriate visuals. If you are presenting sounds as evidence, the ultimate example of coordinating sounds with graphics is a video clip, where the audience sees the sources of the sounds.

- If the sounds are discrete, such as the sounds different animals make, show a photo of each object, and click on the graphic to present the corresponding sound (assuming that this is relevant to your message).

8. Don't arbitrarily vary tonal quality or volume.

If you use sounds, ensure that they all have the same tonal quality and are presented at the same volume—unless changing the quality or volume is part of your message. Any perceptible change will be taken to signal information, and thus arbitrary changes will be confusing and distracting.

9. Ensure that sounds can be heard clearly throughout the room.

As usual, the material must be discriminable, and the sounds must be loud enough to be heard by all.

- In addition, don't make the sounds too loud, which will not only cause the audience members to cringe, but may lead them to anticipate further assaults and thus not pay close attention to your message.

10. Ensure that sounds are high fidelity.

A common problem with presenting sounds is that most computers have poor speakers, so that the sounds can be muddy and difficult to understand. Avoid this problem by ensuring that you have good external speakers, which should have independent amplifiers.

From the Principles to the Point

The key to these recommendations is the psychological principles. In what follows I revisit the principles and note which ones contributed primarily to each of the major recommendations I offer.

Principle of Relevance: Communication is most effective when neither too much nor too little information is presented. (We understand and remember a message more easily when the right amount of detail is used to make the point.)

- Use sounds to provide evidence.

Principle of Appropriate Knowledge: Communication requires prior knowledge of pertinent concepts, jargon, and symbols. (We understand and remember a message more easily if it connects to what we already know.)

- Use video clips to illustrate a relevant event.
- Use sounds that are appropriate for the topic and point being made.
- Use sounds to define the context.

Principle of Salience: Attention is drawn to large perceptible differences. (Big relative differences grab attention.)

- Make the most important content element the most salient.
- Use one color for titles and another, less salient one for text.
- Use warm colors to define a foreground.
- Ensure that the background pattern is not salient.
- Use animation to direct attention.
- Use novel transitions to catch attention.
- Use sounds to grab the audience's attention.
- In general, don't use transitions in which slides slowly appear or fade out.

Principle of Discriminability: Two properties must differ by a large enough proportion or they will not be distinguished. (We need contrast to distinguish shapes, colors, or positions from each other and from the background.)

- Use colors that are well separated in the spectrum.
- Avoid using red and blue in adjacent regions.
- Make adjacent colors have different lightnesses.
- Ensure that the foreground and background are discriminable.
- Don't use deep, heavily saturated blue for text or graphics.
- Ensure that textures are discriminable.
- Avoid visual beats.
- Ensure that sounds can be heard clearly throughout the room.
- Ensure that sounds are high fidelity.

Principle of Perceptual Organization: People automatically group elements into units, which they then attend to and remember. (These groups are easier to see and remember than the isolated components would be.)

- Use color to group elements.

- Don't use hue, lightness, and saturation to specify different measurements.
- Don't allow parts of background patterns to group with parts of the foreground.
- Vary the orientation of similar texture elements in different regions by at least 30 degrees of tilt.
- Vary the spacing of texture elements with similar orientations in different regions by a ratio of at least two to one.
- Don't move portions of the same text line or graphic separately.

Principle of Compatibility: A message is easiest to understand if its form is compatible with its meaning. (For better or worse, the mind tends to judge a book by its cover.)

- Avoid using hue to represent quantitative information.
- Increase saturation and lightness for hues that indicate greater amounts.
- Respect compatibility and conventions of color.
- Make animations compatible with the represented object or event.
- Use sounds that are appropriate for the topic and point being made.
- Use a background pattern to reinforce the main point of the presentation.
- Coordinate sounds, text, and graphics.

Principle of Informative Changes: People expect changes in properties to carry information. (And we expect every necessary piece of information to be indicated by a change in a perceptible property.)

- Don't randomly use different transitions for different slides.
- Use sounds sparingly as alerts.
- Don't arbitrarily vary tonal quality or volume.

Principle of Capacity Limitations: People have a limited capacity to retain and to process information and so will not understand a message if too much information must be retained or processed. (From a communicative point of view, less can be more!)

- Build up complex displays a part at a time.
- Don't move more than four separate perceptual groups simultaneously.
- Use sounds to allow the audience to "come up for air."
- Don't require viewers to read moving words.